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ShipsMOT: A Comprehensive Benchmark and Framework for Multiobject Tracking of Ships

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ABSTRACT

Multiobject tracking of ships is crucial for various applications, such as maritime security and the development of ship autopilot systems. However, existing ship visual datasets primarily focus on ship detection tasks, lacking a fully open-source dataset for multiobject tracking research. Furthermore, current methods often struggle with extracting appearance features under complex sea conditions, varying scales and different ship types, affecting tracking precision. To address these issues, we propose ShipsMOT, a new benchmark dataset containing 121 video sequences with an average of 15.45 s per sequence, covering 15 distinct ship types and a total of 237,999 annotated bounding boxes. Additionally, we propose JDR-CSTrack, a ship multiobject tracking framework that improves feature extraction at different scales by optimising a joint detection and Re-ID network. JDR-CSTrack utilises the fusion of appearance and motion features for multilevel data association, thereby minimising track loss and ID switches. Experimental results confirm that ShipsMOT can serve as a benchmark for future research in ship multiobject tracking and validate the superiority of the proposed JDR-CSTrack framework. The dataset and code can be found on https://github.com/jpj0916/ShipsMOT.

1 | Introduction

Multiobject tracking (MOT) of ships based on deep learning plays a significant role in civilian and military applications. It can recognise and track illegal ships in military fields, which contributes to maritime security and combat exercises. Meanwhile, it is extensively employed in civilian fields for ship monitoring, maritime rescue and ship autopilot. MOT methods based on deep learning need extensive annotated data for training to extract relevant features effectively. An optimal dataset should include diverse backgrounds, complex maritime environments and a rich variety of ship behaviours to improve the robustness and generalisation capabilities of the system. Although existing ship detection datasets such as SeaShips [1], McShips [2], GLSD [3] and SMD [4] have advanced the field of computer vision, they are hindered by several limitations, including a narrow range of ship categories and backgrounds, limited data volume etc. Moreover, they are not directly applicable to MOT tasks. Regarding existing datasets for ship tracking, MarDCT [5] was collected by a surveillance system and contains only nine videos of ships with low resolution. LMD-TShip [6] is more extensive, containing 191 video sequences with an average of 212 frames per sequence. However, it has fewer objects in each video sequence and the dataset is not easily available, which limits its application in multiobject tracking of ships. Therefore, there remains a pressing need for a

Fang Luo, Pengju Jiang and Wenjing Zeng contributed equally to this work.

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ship dataset with rich variations, more ship objects and public availability.

In this paper, we build a ships multiobject tracking dataset named ShipsMOT, which encompasses both the intraclass and interclass variations of ships during sailing, such as categories, lighting conditions, backgrounds etc., and it also considers some real-world challenges encountered by ships while sailing, including occlusions and object similarities. Furthermore, the dataset encompasses 15 ship categories and an extensive collection of over 50,000 frames and 230,000 bounding boxes. In terms of quantity and scope, it stands as the most substantial dataset of its kind. The diversity and volume of the data offer a more accurate representation of ship objects in real-world ship sailing scenarios, thereby setting a benchmark for MOT research in ships.

Additionally, ship multiobject tracking faces several challenges. On one hand, environmental factors such as weather changes and ship occlusions increase the difficulty of trajectory prediction. On the other hand, the variability in size and shape among ships impedes trackers' ability to track objects across different scales. Detection-based tracking methods (tracking by detection) [7–9] are widely used for their flexibility, not relying on a single detection or tracking model. However, their tracking results highly depend on detection performance. Complex environmental conditions combined with unique ship movement patterns can lead to incomplete or distorted object features, which affect detection and tracking performance. Transformer-based methods [10, 11] excel in global feature extraction but require substantial computational resources and training samples, which limits their use in real-time tracking. Joint detection and Re-ID methods [12–14] address these issues by simultaneously outputting bounding boxes and appearance information within the same network, which reduces the impact of detection errors and enhances real-time performance. However, existing trackers are primarily designed for general objects without fully accounting for ship-specific characteristics, which may introduce learning conflicts between detection and Re-ID tasks, thereby compromising the accuracy of ship multiobject tracking. To address this issue, we propose the JDR-CSTrack framework for ship multiobject tracking. This framework integrates a ship object detection module (joint detection and Re-ID for YOLO, JDR-YOLO) and a ship object association module (confidence-based ship multiobject tracking, CSTrack). It enhances the compatibility and coordination between object detection and Re-ID feature extraction and employs a confidence-based adaptive feature fusion (CAFF) strategy combined with multilevel data association. This approach effectively minimises ID switches and track losses, thereby improving overall tracking performance.

In summary, our main contributions are as follows:

- We build a new ships multiobject tracking (ShipsMOT) dataset. It contains various movement patterns of ships, as well as video data of ships from varying backgrounds and different real-world conditions. We have made the dataset fully open-source, which offers data support for MOT research in ships.
- We propose a multiobject tracking framework, designed for the characteristics of ship objects. It captures multiscale feature information during the feature extraction phase and performs multilevel data association based on the fusion features of appearance and motion, thereby enhancing the accuracy of ship multiobject tracking within complex environments.
- We conduct a comprehensive analysis of both our dataset and the proposed multiobject tracking framework. Experimental results demonstrate the effectiveness of our dataset and the superiority of the proposed framework.

2 | Related Work

2.1 | Multiobject Tracking Datasets

In the field of MOT, existing datasets focus on tracking pedestrians and vehicles, such as MOT15 [15], MOT16 [16], MOT17 and MOT20 [17], which are designed for pedestrian tracking in high-density object scenarios. The KITTI tracking [18] dataset focuses on tracking vehicle traffic across a variety of driving environments, including urban, rural and highway settings.

Moreover, some MOT datasets have expanded into the field of action recognition and a wider range of objects. For example, SoccerNet-Tracking [19] focuses on soccer videos and contains 200 tracking sequences covering 11 soccer action categories (e.g., goals, corners, direct free kicks, fouls etc.). SportsMOT [20] is a large-scale MOT dataset for multiple sports backgrounds such as soccer, volleyball and basketball fields; its advent fills the gap for sports background benchmarks in MOT. DanceTrack [21] specialises in tracking objects with similar appearances but complex movements, including group dances, martial arts and gymnastics. IS-MOT [22] instead focuses on the fine-grained postures of indoor pedestrians, with greater emphasis on fine-grained actions such as head movements, interactive actions and body movements. TAO [23] is a COCO [24]-like diverse category MOT dataset, aiming to achieve tracking of any object in the real world.

Despite the existence of these diverse datasets, there remains a scarcity of MOT datasets that specifically focus on ship objects. To address this limitation, we build the ShipsMOT dataset for multiobject tracking of ships.

2.2 | Multiobject Tracking Methods Based on Joint Detection and Re-ID

Joint detection and Re-ID methods are widely employed in multiobject tracking (MOT) frameworks. These methods typically perform object detection on each frame to localise candidate objects within bounding boxes. Subsequently, they generate a similarity matrix by extracting features from the bounding boxes. Data association techniques, such as the Hungarian algorithm or greedy algorithms, are then applied to perform object tracking.

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SORT [7] integrates Kalman filtering with the Hungarian algorithm to achieve real-time object tracking. However, traditional Kalman filtering may not accurately predict the nonlinear movements of ships, such as avoiding obstacles and turning, which potentially impacts tracking stability. DeepSORT [8] enhances the performance of SORT by introducing a Re-ID network but increases processing time. JDE [12] further optimises DeepSORT by integrating object detection and Re-ID feature learning within the same network. However, JDE employs an anchor-based object detection model, which may result in overlapping objects at each location, thereby impeding effective embedding learning. FairMOT [13] addresses this issue by employing an anchor-free object detection method and predicting the centres and bounding boxes of objects on high-resolution feature maps, which avoids the ambiguities and alignment issues of anchor-based methods. However, it may still struggle with adequately handling occlusions and may experience a decline in tracking performance in complex backgrounds.

In complex maritime environments, such as varying weather conditions, ship occlusions and significant variations in ship scales, there is a risk of misclassifying genuine objects as background during information processing across different feature layers. Consequently, the main challenges in the field of ship multiobject tracking involve balancing feature utilisation between detection and Re-ID tasks, as well as enhancing tracking accuracy and stability under complex maritime conditions.

3 | ShipsMOT

Ship detection and multiobject tracking technologies are crucial in both military and civilian applications. Existing research indicates a lack of publicly available large-scale datasets specifically designed for ship multiobject tracking (MOT). This paper aims to establish a benchmark for ship multiobject tracking research by building a novel dataset—ShipsMOT.

3.1 | Dataset Construction

To build the ShipsMOT dataset, we collected a portion of the videos from live maritime surveillance systems. Furthermore, to enhance the variety of ship trajectories in the dataset, we also collected videos from the Internet. Each video in the dataset has a resolution of 1920 × 1080 or 1280 × 720, with durations ranging from several tens of seconds to a few minutes. The criteria for video selection encompassed public availability, a higher number of ship trajectories and various backgrounds.

We used DarkLabel to annotate the collected video sequences in the MOT format, a specialised format designed to precisely capture the movement trajectories and states of multiple objects within video sequences. In the MOT format, each frame's data contains nine distinct fields, the details of which are provided in Table 1. The annotation process adhered to the following principles: (1) We annotated objects that could be inferred as ships through previous frames; (2) we chose to keep the same ID for an object that leaves the camera frame and comes back at a later time during the same clip; (3) we annotated full-body boxes for partially occluded objects. To ensure the high quality of the dataset, each bounding box has been subjected to automatic tracking by software and manual verification.

TABLE 1 | The data format of MOT.

Position	Name	Description
1	Frame	The number of frames
2	Id	The number of tracks
3	Bb_left	The coordinates of the upper left corner of the bounding box and the length and width of the object
4	Bb_top	
5	Bb_width	
6	Bb_height	
7	Conf	The confidence level
8	X	The content used in MOT3D is always −1 in 2D detection.
9	Y	

3.2 | Dataset Variety

A good MOT dataset should contain several factors that can affect the performance of multiobject tracking. The ShipsMOT dataset integrates varied factors that may affect the performance of ship multiobject tracking, including category, background, lighting condition, viewpoint and scale. Moreover, ShipsMOT also considers real-world challenges encountered by ships during sailing, including occlusions, similar objects and object deformations, which enhance the performance and reliability of the model, making it more effective for practical use.

3.2.1 | Category Variation

We construct 15 categories that widely exist in real-world ship sailing scenarios. The aim is to help the model refine its comprehension of the general features and attributes of ships by identifying and distinguishing these categories, thereby facilitating more precise detection and tracking in practical applications (examples in Figure 1).

3.2.2 | Background Variation

ShipsMOT contains six distinct water environments such as city, coast, harbour etc. The aim is to capture the sailing status of ships in various real-world water environments. This ensures that the model is comprehensively trained in each scenario, thereby enhancing its generalisation capabilities and practical applicability (examples in Figure 2).

3.2.3 | Illumination Variation

Lighting conditions profoundly affect ship detection and tracking. To ensure robust performance in diverse lighting conditions, we enrich the dataset with object images captured under varying illumination states, including sunny days, cloudy

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days and the lighting transitions at sunrise, sunset and night-time. Figure 3a shows three images depicting distinct lighting conditions. From left to right, they present scenes of daytime, dusk and night-time, respectively. These diverse scenarios enable the model to adapt to object detection tasks under different lighting conditions, enhancing its generalisability.

3.2.4 | Viewpoint Variation

ShipsMOT incorporates a range of viewpoints for ship activity backgrounds, such as aerial, side-on and front-facing perspectives, along with observations taken from different positions and altitudes. Figure 3b, from left to right, respectively shows ship-related video images captured from the bow-standing viewpoint, the overhead (top-down) view and the side-viewing angle. These provide a rich diversity of viewpoints to help the model adapt to object detection and tracking tasks under varied observational stances.

3.2.5 | Scale Variation

ShipsMOT encompasses ships of diverse scales, ranging from large passenger ships and oil carriers to smaller crafts such as canoes and kayaks. It also contains ships of the same type with significant scale differences caused by varying distances from the camera. Figure 3c presents representative video images of ships with distinct scale variations. As shown, the largest ship occupies roughly one third of the entire frame, whereas some

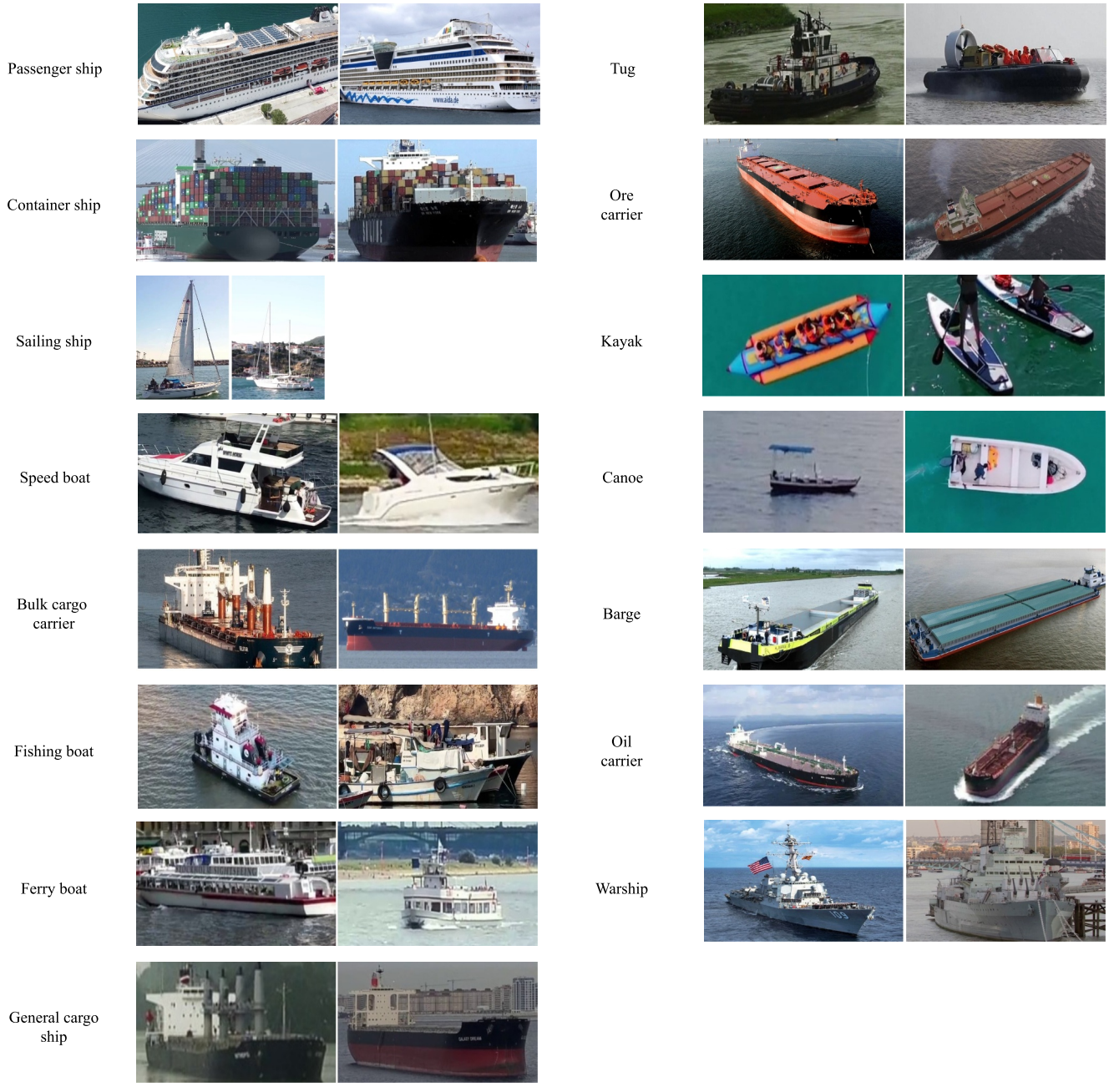


FIGURE 1 | Example images depicting various ship categories.

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FIGURE 2 | Example images depicting various backgrounds.

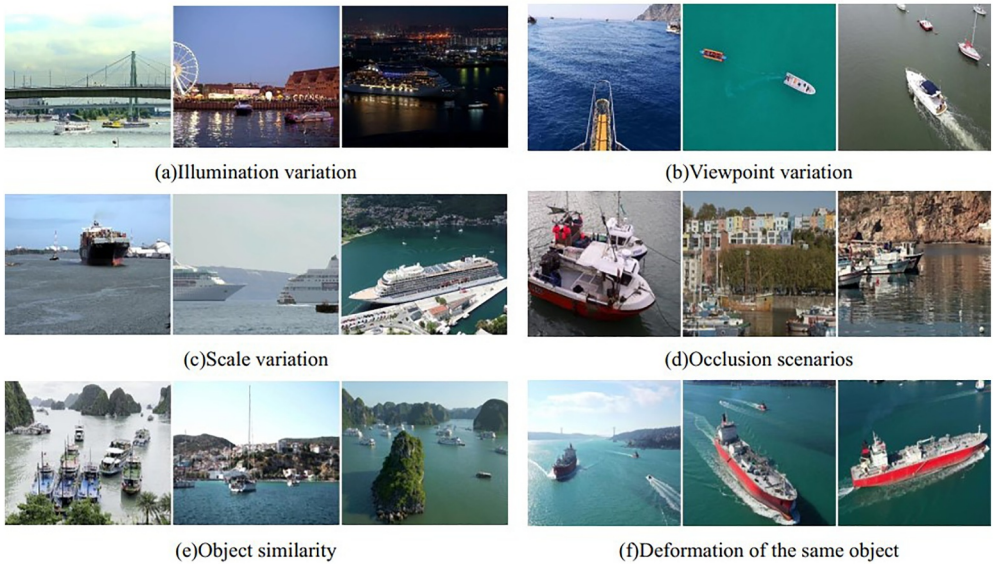


FIGURE 3 | Example images depicting various visual variations.

small ships measure only about 20 × 16 pixels. This scale diversity enables the model to learn features associated with changes across different ship sizes, enhancing its capability to detect and track ships of different scales.

3.2.6 | Occlusion Scenarios

In marine environments, occlusions frequently occur as ships navigate or dock, with numerous occlusion scenarios affecting the accuracy of detection and multiobject tracking. To assess the performance of tracking methods in handling occlusions, we incorporate various occlusion conditions into ShipsMOT, where more than 40% of the video sequences include occlusion scenarios ranging from no occlusion to complete occlusion. Figure 3d shows ship video images with different levels of occlusion, illustrating partial or complete occlusion of specific ships by other ships or buildings.

3.2.7 | Object Similarity

Although significant differences exist between different ship types, the appearance of ships within the same type can be highly similar, presenting substantial challenges for ship tracking. Therefore, ShipsMOT contains numerous videos of ships with similar appearances to enhance the Re-ID performance of tracking models. Figure 3e shows several real-world scenarios, such as ships sailing or berthing, where ships have similar appearances but belong to different categories.

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3.2.8 | Deformation of the Same Object

In real-world sailing environments, the same object can undergo deformation due to changes in distance and viewing angles. ShipsMOT contains videos capturing such scenarios, featuring scale and viewpoint variations of the same object. As shown in Figure 3f, when the red ship is sailing normally, its appearance characteristics present varying degrees of deformation during movement due to the fixed camera lens. This is a critical factor to consider for multiobject ship tracking tasks, as it challenges the model's ability to maintain consistent object association despite visual changes.

3.3 | Dataset Statistics

3.3.1 | Dataset Split

We collected 121 videos in the ShipsMOT dataset, with 81 used for training and 40 reserved for testing. The annotations for both the training and testing sets are publicly available for research purposes. Detailed statistics and data splits for ShipsMOT are provided in Table 2.

3.3.2 | Data Classification

Table 3 lists the number of bounding boxes corresponding to each ship category in ShipsMOT. The varied category distribution and a considerable number of training samples help the model to comprehensively learn and understand ship features, which would enhance the precision and generalisation capabilities of the model in ship detection and MOT tasks.

3.3.3 | Appearance Features

We performed a statistical analysis on the aspect ratios of bounding boxes in ShipsMOT, as shown in Figure 4. The majority of bounding boxes fall between 1 and 4, which corresponds to the shape of ship objects. However, there are also bounding boxes with aspect ratios < 1 or > 4, which indicates the existence of exceptionally long or wide ships. This statistical result shows that ShipsMOT encompasses a wide range of ship types and provides annotations that are finely aligned with the distinctive visual features of each ship.

Moreover, object scales can vary considerably due to changes in distance and viewpoint during ship navigation. To characterise this phenomenon, we performed statistical analysis on the pixel areas of objects in ShipsMOT. Adhering to the object scale classification from the COCO dataset, we categorised the objects into three size categories, as detailed in Table 4. Given the large physical size of ships and their frequent occupation of a significant portion of the video frame during navigation, medium and large objects constitute a considerable proportion. Nevertheless, ShipsMOT also includes a sufficient number of small objects to ensure the model can accurately handle objects of varying sizes after training.

TABLE 2 | Detailed statistics and data splits for ShipsMOT.

Split	Category	Sequences	Avg. len. (s)	Tracklets	Bounding boxes
Train	15	81	16.5	457	185,128
Test	15	40	14.4	151	52,871
Total	15	121	15.45	608	237,999

3.4 | Comparison With Existing Datasets

Compared with existing ship detection or tracking datasets, as shown in Table 5, ShipsMOT supports both object detection and multiobject tracking (MOT). In contrast, datasets such as SeaShips, McShips, GLSD and SMD are limited to object detection only. Although MarDCT and LMD-TShip can be used for tracking tasks, MarDCT is limited to single-object tracking, whereas LMD-TShip supports MOT but cannot be used for object detection; we also observed that most of its annotations

TABLE 3 | Number of bounding boxes for each ship category in ShipsMOT.

Category	Bounding boxes
Passenger ship	58,848
Container ship	13,917
Sailing ship	20,174
Speedboat	21,589
Barge	1734
Fishing boat	26,824
Ferry boat	15,490
General cargo ship	15,506
Oil carrier	3862
Canoe	22,610
Tug	8229
Warship	426
Kayak	1009
Ore carrier	8591
Bulk cargo carrier	19,190

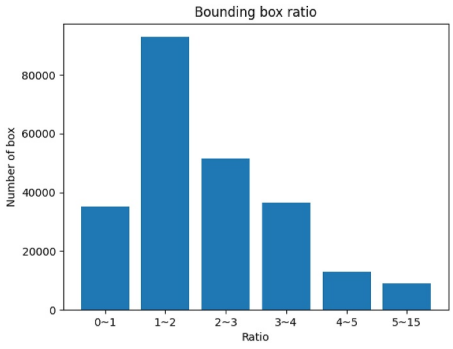


FIGURE 4 | Statistics of bounding box aspect ratios in the ShipsMOT dataset.

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are for single objects. Furthermore, our ShipsMOT classifies ships into 15 categories, includes over 56,000 labelled images, features an average of 5 distinct objects per video sequence and contains more distinctive information than other datasets.

Compared with existing MOT datasets, as shown in Table 6, MOT17 and MOT20 focus primarily on dense pedestrian tracking and contain more trajectories; KITTI is a MOT dataset for autonomous driving; TAO is a general-domain MOT dataset that provides a large number of annotated categories and video sequences; DanceTrack is designed for dancer pose estimation and focuses on more fine-grained features of the objects. Given the research community's current lack of a fully public dataset for ship multiobject tracking, we identified this gap and constructed the ShipsMOT dataset. ShipsMOT is specifically designed for ship multiobject tracking, containing 15 ship categories. It comprises 121 video sequences, with an average of 5 trajectories per video, and covers diverse ship sailing scenarios such as harbour, offshore and coast. Additionally, it contains ship objects of different scales and provides 237,999 annotated bounding boxes, which helps enhance the generalisation capabilities of models. Moreover, ShipsMOT mitigates the scarcity of publicly available datasets for ship multiobject tracking.

TABLE 4 | Distribution of ship objects across different scales in ShipsMOT.

Scale	Pixel area	Number of objects
Small	$\leq 32 \times 32$	32,426
Medium	$> 32 \times 32, \leq 96 \times 96$	125,469
Large	$> 96 \times 96$	80,104

TABLE 5 | Comparison of statistics between existing ship detection or tracking datasets and the ShipsMOT dataset.

Dataset	Task	Category	Sequences	Total images
SeaShips	Detection	6	—	7000
McShips	Detection	2	—	8000
GLSD	Detection	13	—	100,729
SMD	Detection	6	81	31,653
MarDCT	SOT	N/A	9	23,200
LMD-TShip	MOT	5	191	40,240
ShipsMOT	Detection/MOT	15	121	56,568

4 | JDR-CSTrack

In ship multiobject tracking tasks, the main challenges arise from inadequate recognition of object appearances and scale variations, leading to inaccurate feature extraction. Furthermore, complex environments often cause the discontinuity or loss of ship trajectories, thereby impacting data association. To address these issues, we propose the JDR-CSTrack framework, which offers a more efficient and precise solution for ship multiobject tracking.

4.1 | Overview

JDR-CSTrack is a MOT framework based on joint detection and Re-ID network. It consists of two modules: object detection and object association; the overall framework is shown in Figure 5. In this work, the input video sequence is first processed by JDR-YOLO to obtain the object detection boxes and their corresponding appearance features (Re-ID features) for the current frame. CSTrack is employed for object association, which initially extracts existing ship trajectories and motion features via Kalman filter. Subsequently, proceeding to the stage of multilevel data association between detection boxes and trajectories, the first and second associations employ our proposed confidence-based adaptive feature fusion (CAFF) strategy. It utilises fused feature similarity, which is calculated from appearance feature similarity and motion feature similarity, to guide the association of high-score and mid-score bounding boxes. The third association uses IoU to associate low-score bounding boxes.

4.2 | JDR-YOLO

The JDR-YOLO model, based on YOLOv8 [25], primarily consists of three parts: backbone, neck network and detection head. Figure 6 shows the structure of JDR-YOLO. The backbone employs a concatenated structure of CBS, multiCSP and SPPF modules to extract features from the input video frames. The neck network fuses the various feature layers from the backbone. To enhance the recognition capability of ship objects in complex scenes, we add the Re-ID feature extraction branches to the detection head. Additionally, the RFB-CA module is added before each detection head to broaden its receptive field and more accurately capture ship features. Furthermore, to ensure objects of varying scales receive adequate attention, we use the dynamic focusing loss function WIoU [26] to coordinate learning tasks.

TABLE 6 | Comparison of statistics between existing MOT datasets and the ShipsMOT dataset.

Dataset	MOT17	MOT20	KITTI tracking	TAO	DanceTrack	ShipsMOT
Domain	Pedestrians	Urban (crowded)	Autonomous driving	Generic	Motion	Ships
Category	12	5	2	833	1	15
Sequences	14	8	50	2907	100	121
Avg. len. (s)	35.4	66.8	12.6	36.8	52.9	15.4
Avg. tracklets	91	432	19	5	10	5
Bounding boxes	292,733	2,102,385	65,213	332,401	877,000	237,999

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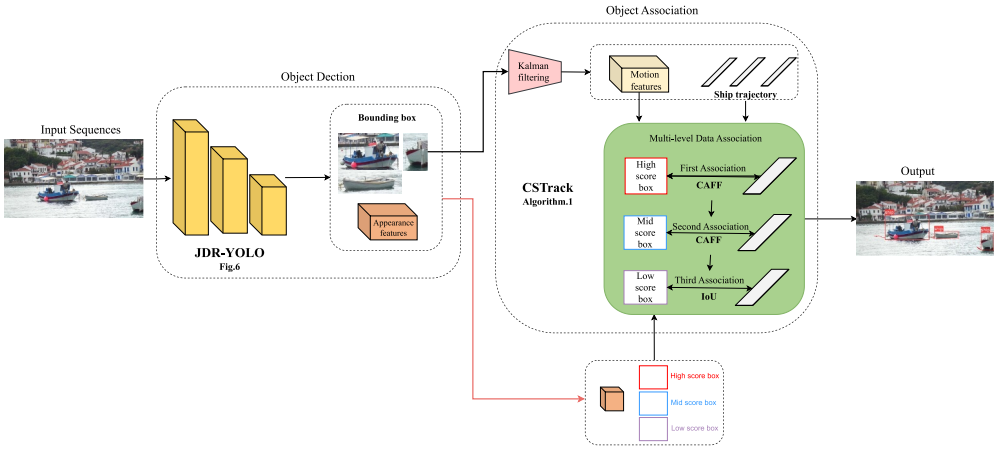


FIGURE 5 | The framework of the proposed JDR-CSTrack.

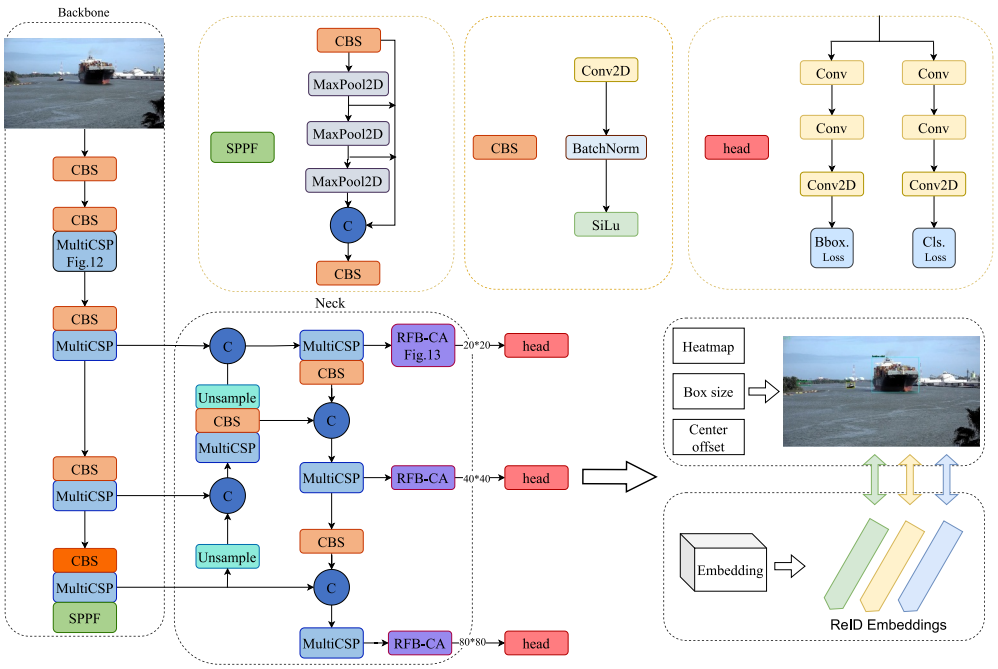


FIGURE 6 | The architecture of the JDR-YOLO model.

4.2.1 | MultiCSP

YOLOv8 utilises CSPDarkNet53 [27] as its backbone. CSPDarkNet53 enhances the learning capability of neural networks while reducing computational complexity, which provides more efficient and accurate feature extraction. However, directly employing CSPDarkNet53 in a joint detection and Re-ID network may result in learning conflicts between object detection and Re-ID tasks. Re-ID tasks require shallow features to discriminate objects, whereas object detection tasks rely on deeper semantic features from the network. To address this issue, we designed the multiCSP structure, which introduces branches and cross-layer connections to ensure effective utilisation of features at various levels. Furthermore, to optimise parallel computing across various branches, we employ Rep convolutional layers [28] in place of residual blocks. The Rep convolutional layers adopt a multibranch structure during training to optimise the network training process and integrate the branches during inference. This expands the width of the network and amplifies its feature extraction capabilities while also enhancing the efficiency of parallel computing.

Figure 7 shows the structure of multiCSP, in which the input image first undergoes a 3×3 convolution to extract initial features. After this, the image features are processed by two 1×1 convolutional layers, which divide the features into two distinct branches. One of these branches' feature maps passes through two stacks of n Rep convolutional layers to obtain two outputs. These outputs are then concatenated with the previous two feature maps to obtain an integrated feature map. This concatenated feature map undergoes a 1×1 convolutional layer to regain its original channel count.

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4.2.2 | RFB-CA

A major challenge in multiobject tracking of ships arises from the scale variations of the objects, which hinder the accurate extraction of appearance features. When objects are excessively large, the receptive field may fail to encompass the entire ship, and for smaller objects, it may inadvertently capture excessive background details. Simply increasing or decreasing the receptive field can either introduce background interference or extract incomplete features. To address this issue, we designed the RFB-CA module, which utilises dilated convolutions with dilation rates of 1, 3 and 5 to broaden the receptive field. Additionally, we introduce the CA [29] attention module, which focuses on the feature layer in both channel and spatial dimensions. This strategy enables the network to prioritise authentic object information while mitigating the influence of background information.

Figure 8 shows the structure of the RFB-CA module, in which the input feature layer of size H × W × C is processed through three convolutional branches with distinct dilation rates to obtain an output feature layer × of size H × W × 3C. After this, the CA attention module splits the feature map into two directions: width and height. Each direction undergoes global average pooling to obtain two distinct feature maps. The calculation for this process is defined in Equations (1) and (2):

$$Z_c^h(h) = \frac{1}{W} \sum_{0 \le i < W} x_c(h, i) \tag{1}$$

$$Z_c^w(w) = \frac{1}{H} \sum_{0 \le j < H} x_c(j, w) \tag{2}$$

The generated feature maps of size 3C × 1 × W and 3C × H × 1 are transformed and then concatenated. Subsequently, a 1 × 1

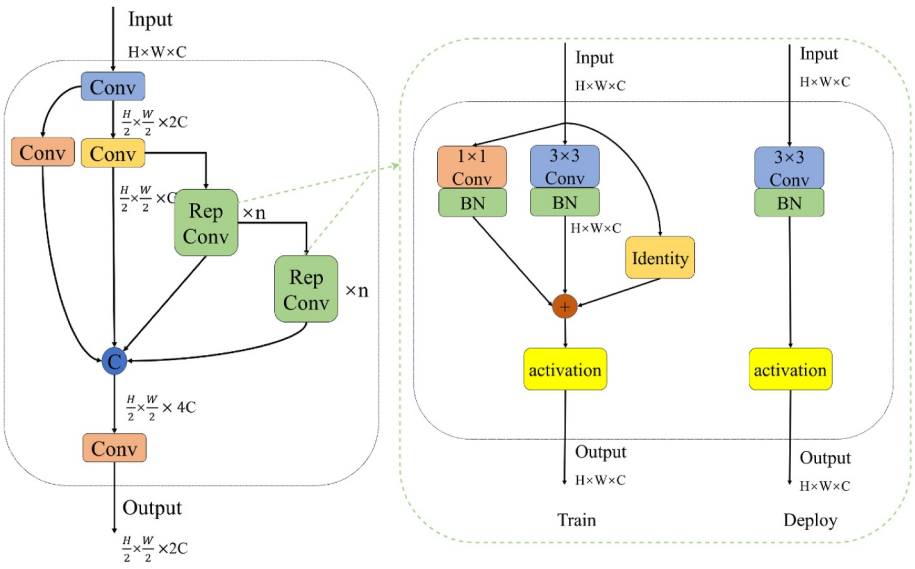


FIGURE 7 | The structure of the multiCSP module.

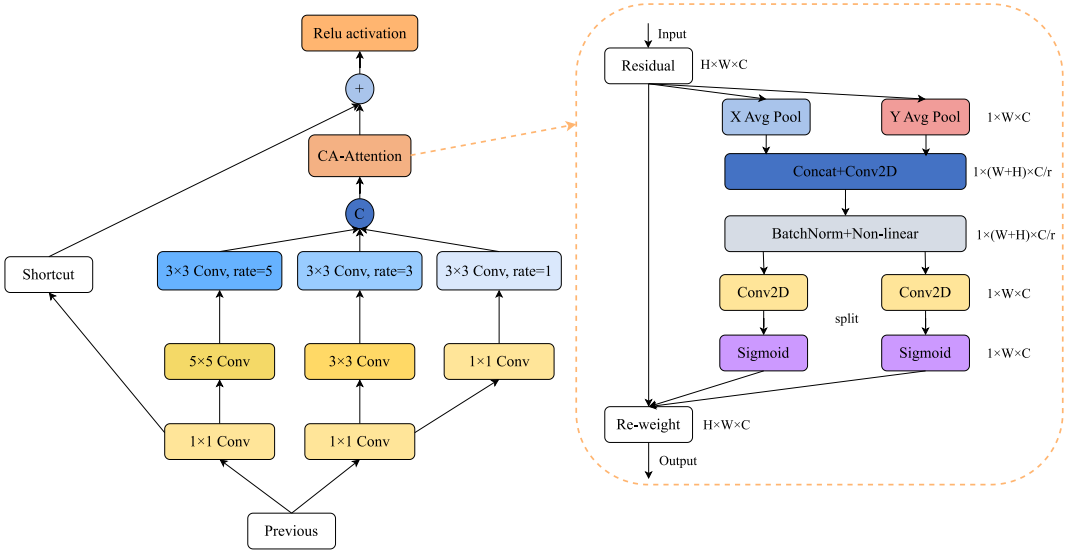


FIGURE 8 | The structure of the RFB-CA module.

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convolution operation F_1 is applied to these concatenated maps, which reduces their dimensionality to obtain the feature map with dimensions 3C/r × × (H + W) × 1. The calculation for this process is defined in Equation (3):

$$f = Sigmoid\left(\left(F_1\left[z^h, z^w\right]\right)\right) \tag{3}$$

Subsequently, the feature map f undergoes a split operation along the spatial dimensions, dividing it into two distinct parts: f^h with dimensions 3C/r × H × × 1 and f^w with dimensions 3C/r × 1 × × W. Each of these parts is individually processed through 1 × 1 convolutions to enhance their dimensionality, followed by the application of the activation function to obtain the final attention vectors g^h and g^w. The calculation for this process is defined in Equations (4) and (5):

$$g^h = Sigmoid\left(F_h\left(f^h\right)\right) \tag{4}$$

$$g^w = Sigmoid\left(F_w\left(f^w\right)\right) \tag{5}$$

The output of the CA attention module is defined in Equation (6):

$$y_c(i,j) = x_c(i,j) \times g_c^h(i) \times g_c^w(j) \tag{6}$$

Finally, a 1 × 1 convolution is employed to condense the number of channels to C and is summed with the cross-layer connection to obtain the final output.

4.2.3 | Dynamic Focusing Loss Function

The traditional IoU metric primarily focuses on the overlapping portion between predicted and ground truth boxes while disregarding the nonoverlapping areas, which may lead to biases in evaluation. To address this issue, we use the dynamic focusing loss function (WIoU). WIoU first calculates the IoU between predicted and ground truth boxes and then assigns a weight factor based on the scale characteristics of different objects. This weight factor adjusts the IoU calculation, which ensures that matching accuracy remains not compromised by variations in object scale and shape. WIoU is defined in Equations (7–9):

$$L_{WIoU} = \frac{\beta}{\delta\alpha^{\beta-\delta}} R_{WIoU} L_{IoU} \tag{7}$$

$$\beta = \frac{L_{IoU}^*}{\overline{L_{IoU}}} \in [0, \ +\infty) \tag{8}$$

$$R_{WIoU} = \exp\left(\frac{\left(x - x_{gt}\right)^2 + \left(y - y_{gt}\right)^2}{\left(W_g^2 - H_g^2\right)^*}\right) \in [1, e) \tag{9}$$

Where β is the weight factor, and W_g and H_g are the dimensions of the smallest bounding box. To prevent R_{WIoU} from generating gradients that might impede convergence, W_g and H_g are detached from the computation graph (the superscript $*$ indicates this operation), which ensures that these dimensions do not contribute to gradient calculation and propagation.

4.3 | CSTrack

In real-world sailing environments such as ports, the presence of numerous ships and frequent occlusions can lead to the detection confidence of some authentic objects falling below the detection threshold, which results in missed detections and interrupted trajectories. ByteTrack effectively mitigates the challenges of trajectory disruptions due to occlusions by utilising low-confidence bounding boxes for secondary matching. However, ByteTrack only utilises motion features during tracking, which describe the temporal motion state of the objects and aid tracking methods in predicting future positions. In scenarios with dense, similar objects and complex motion patterns, the tracking performance of ByteTrack may be lowered. In such cases, researchers can significantly improve object recognition in occluded or dense scenes by incorporating Re-ID features that capture object appearance characteristics. However, traditional methods for fusing motion and appearance features often rely on a fixed weighted sum approach. During the object association stage, the reliability of appearance features may decrease when objects are occluded or small. In such cases, prioritising motion states such as position and velocity can substantially enhance tracking accuracy. We observe that the confidence of a detection box can serve as an indicator of the reliability of its appearance information. As shown in Figure 9, the lower the confidence of a bounding box, the more blurred its appearance features tend to be.

In this paper, we propose the confidence-based adaptive feature fusion (CAFF) strategy, which dynamically adjusts weights to address scenarios where appearance features may be less dependable. It operates by increasing the weight of motion features for detection boxes with lower confidence and increasing the weight of appearance features for those with higher confidence. This enables the algorithm to capitalise on the most relevant feature information, thereby enhancing the accuracy and robustness of multiobject tracking. The fusion method is defined in Equation (10):

$$S = e^{-c} S_i + (1 - e^{-c}) S_j \tag{10}$$

Where S represents the fused feature similarity, S_i and S_j refer to the similarity between the detection box and the features of motion and appearance, respectively, and c represents the confidence score of the detection box. Additionally, ByteTrack only distinguishes between high- and low-confidence detection boxes, which may lack flexibility in different scenarios and object densities. In this paper, we categorise detection boxes into three levels based on confidence: high, medium and low, and then perform multilevel data association between bounding boxes and trajectories. The pseudocode of multilevel data association is shown in Algorithm 1.

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FIGURE 9 | Examples of detection boxes with varying confidence levels.

ALGORITHM 1 | Pseudocode of multilevel data association.

Require: A video sequence V; detection boxes D; confidence scores of detection boxes *d.score*
Ensure: Tracks T of the video
Hyper parameters: detection score threshold $t_{high} = 0.6$, $t_{mid} = 0.4$, $t_{low} = 0.2$
Initialisation $T \leftarrow \varnothing$
for frame f_k in V **do**
  // Classify detection boxes based on confidence
  $D_{low} \leftarrow \varnothing$; $D_{mid} \leftarrow \varnothing$; $D_{high} \leftarrow \varnothing$
  for *d.score* in D **do**
    if $d.score > t_{high}$ **then**
      $D_{high} \leftarrow D_{high} \cup d$
    else if $d.score > t_{mid}$ **then**
      $D_{mid} \leftarrow D_{mid} \cup d$
    else if $d.score > t_{low}$ **then**
      $D_{low} \leftarrow D_{low} \cup d$
    end if
  end for
  for t in T **do**
    // Use Kalman filter to predict tracks
    $\hat{x} \leftarrow F \cdot x_t$ // State prediction, x_t is previous state estimate
    $\hat{P} \leftarrow F \cdot P_t \cdot F^T + Q$ // Covariance prediction
    // Update step
    if there exists a detection d associated with track t **then**
      $z \leftarrow$ state vector from detection d (position, velocity etc.)
      $y \leftarrow z - H \cdot \hat{x}$
      $S \leftarrow H \cdot \hat{P} \cdot H^T + R$
      $K \leftarrow \hat{P} \cdot H^T \cdot S^{-1}$ // Kalman gain
      $x' \leftarrow \hat{x} + K \cdot y$ // State update
      $P' \leftarrow (I - K \cdot H) \cdot \hat{P}$ // Covariance update
      $t \leftarrow (x', P', z, H, F, Q, R)$ // Update track state
    else
      $t \leftarrow (\hat{x}, \hat{P}, z, H, F, Q, R)$ // No update, use prediction only
    end if
  end for
  KM$(D_{high}, T) \rightarrow T_1, D_1$ // First data association: high confidence
  KM$(D_{mid}, T_1) \rightarrow T_2, D_2$ // Second data association: medium confidence
  KM$(D_{low}, T_2) \rightarrow T_{lost}$ // Third data association: low confidence
  $T \leftarrow T \backslash T_{lost}$ // Remove lost tracks
  $T \leftarrow T \cup \text{init}(D_1, D_2)$ // Initialise new tracks
end for
Return T

The main flow of the multilevel data association is as follows:

1. Categorise detection boxes: The detector performs detection on the current frame, taking object detection boxes D and their corresponding confidence scores *d.score* as input. By contrasting the confidence scores against the detection box thresholds, the boxes are divided into high-confidence boxes D_{high}, medium-confidence boxes D_{mid} and low-confidence boxesD_{low}.

2. Predict tracks: Adopt Kalman filtering to predict the new locations in the current frame of each track in T (including activated and deactivated tracks, with deactivated tracks only added to the activated tracks after successfully matching two or more consecutive frames). If predicting tracks for the first frame, because T is empty at this point, the Kalman filtering initialises the state vector (position/velocity) using the first detected bounding box. In subsequent frames, it predicts the next state using a dynamic model (e.g., constant velocity), then updates this prediction by fusing it with new detections (if available), weighing both on their uncertainties to minimise error. This recursive cycle enables robust trajectory estimation, handling occlusions by relying on predicted states when detections are missing and refining tracks with each new observation, ensuring accurate long-term tracking.

3. First data association: Associate the high-confidence boxes D_{high} with all tracks in T by using the CAFF strategy. For successfully matched tracks, update their appearance and motion features; unmatched tracks are saved as T_1.

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4. Second data association: Associate the medium-confidence boxes D_{mid} with the unmatched tracked tracks in T_1 that have been missing for 30 frames or less by using the CAFF strategy. For successfully matched tracks, update their appearance and motion features; unmatched tracks are saved as T_2.
5. Third data association: Match the low-confidence boxes D_{low} with the tracked tracks in T_2 by using IoU. For successfully matched tracks, update their appearance and motion features; unmatched low-confidence boxes are directly discarded; the remaining unmatched tracks are saved as T_{lost} and remove unmatched tracks that have been missing for more than 50 frames in T_{lost}.
6. Update tracks: Create new tracked tracks from the unmatched high-confidence detection boxes after the first association; create new deactivated tracks from the unmatched medium-confidence detection boxes after the second association; add these newly created tracks and the tracks in T_{lost} to T.
7. Process the next frame: Return to Step (1) for multilevel data association again until all video frames have been processed, at which point the algorithm execution is complete.
8. Output results: The output of each frame is the bounding boxes and ID label of the tracks in the current frame.

5 | Experiments

5.1 | Experiments Setup

5.1.1 | Dataset Configurations

ShipsMOT is designed for ship object detection and ship multiobject tracking. For object detection evaluation, we compare ShipsMOT with existing ship object detection datasets: McShips and SeaShips. For multiobject tracking evaluation, due to the absence of publicly available ship-specific MOT datasets, we performed a comparative analysis between ShipsMOT and widely used MOT datasets, MOT17 and MOT20.

5.1.2 | Evaluation Metrics

For the evaluation of object detection, we use commonly used metrics: recall, precision and mean average precision (mAP). For the evaluation of multiobject tracking performance, our evaluation protocol prioritises MOTA [30] and HOTA [31]. MOTA is used as the main metric for evaluation, but it emphasises detection quality over association quality. HOTA considers the accurate detection, association and localisation of objects. We also use AssA, IDF1 and IDs to emphasise aspects of association performance and DetA for detection performance.

5.1.3 | Implementation Details

In the object detection task, we employed YOLOv8l as the pretrained model. Subsequently, we trained this model on SeaShips, McShips and ShipsMOT for 80 epochs, with a batch size of 16, using a Tesla V100 GPU, a learning rate of 0.01 and warmup epochs of 3, to evaluate the impact of different datasets on model performance. In the MOT task, we evaluated the performance of classic MOT methods on MOT17, MOT20 and ShipsMOT and compared these methods with our proposed JDR-CSTrack. The effectiveness of our improvements was subsequently validated via an ablation study.

5.2 | Results on ShipsMOT Dataset

5.2.1 | The Task of Object Detection

Table 7 shows the cross-dataset detection results. Models trained on ShipsMOT outperform those trained on SeaShips and McShips in recall, precision and mAP across both datasets. Specifically, the ShipsMOT-trained model achieves a 5.1% higher mAP50 on SeaShips and a 10.3% higher mAP50 on McShips. Moreover, when tested on ShipsMOT, the mAP of models trained on the other two datasets did not exceed 70%. The reason is that ShipsMOT contains more diverse scenes, more ships and ship types compared to the scene simplification in SeaShips and the low object density in McShips, which contributes to helping advance ship object detection performance, demonstrating that the diversity and volume of ShipsMOT not only improve in-domain detection but also enhance cross-domain generalisation.

5.2.2 | The Task of Multiobject Tracking

To validate that the ShipsMOT dataset can be set as a benchmark for ship multiobject tracking tasks, we evaluate several classic MOT methods, including CenterTrack, FairMOT, BoT-SORT and ByteTrack, on MOT17, MOT20 and ShipsMOT, as shown in Table 8. CenterTrack and FairMOT employ CenterNet [32] as detectors, whereas the remaining trackers utilise YOLOv8 or YOLOX [33] (the superscript * indicates the utilisation of YOLOv8). The results show that ShipsMOT achieves consistent performance trends across tracking methods compared to MOT17 and MOT20 (CenterTrack performs the worst and ByteTrack* performs the best). This demonstrates that ShipsMOT has effective evaluation capabilities for distinguishing the performance of various methods. However, it is noteworthy that the MOTA for each tracking method on ShipsMOT is lower compared to other datasets. The reason is the significant scale variations and appearance differences between ship objects and pedestrian objects. Consequently, we propose

TABLE 7 | Ship detection results on different training sets.

Train	Test	Precision	Recall	mAP50
SeaShips	ShipsMOT	0.643	0.747	0.629
	McShips	0.763	0.663	0.632
McShips	ShipsMOT	0.71	0.587	0.623
	SeaShips	0.783	0.656	0.711
ShipsMOT	McShips	0.758	0.678	0.735
	SeaShips	0.899	0.665	0.762

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the JDR-CSTrack framework in this paper, which is specifically designed for ship multiobject tracking.

5.3 | Results on JDR-CSTrack Framework

5.3.1 | The Task of Object Detection

To validate the performance of JDR-YOLO for ship object detection, we compare JDR-YOLO with CenterNet, YOLOX and YOLOv8, as shown in Table 9. It can be seen that JDR-YOLO outperforms CenterNet, YOLOX and YOLOv8 by integrating three key innovations: (1) multiCSP structure for conflict-free feature learning between detection and Re-ID, (2) RFB-CA module to adaptively expand receptive fields for scale-varying ships and (3) WIoU loss to optimise bounding box regression for diverse ship aspect ratios. The model achieves 0.943 mAP50, a 1.1% gain over YOLOv8.

5.3.2 | The Task of Multiobject Tracking

To validate the performance of CSTrack for ship multiobject tracking, we compare CSTrack with CenterTrack, FairMOT, ByteTrack and BoT-SORT, as shown in Table 10. It is observed that the tracking methods employing YOLOX as the detection model outperform CenterTrack and FairMOT in all metrics, with CSTrack achieving the best performance.

TABLE 8 | Tracking performance of various MOT methods on MOT17, MOT20 and ShipsMOT.

Dataset	Method	HOTA↑	MOTA↑	IDF1↑
MOT17	CenterTrack	52.2	67.8	64.3
	FairMOT	59.3	73.3	72.3
	ByteTrack	63.1	80.3	77.3
	BoT-SORT	64.6	80.6	79.5
	ByteTrack*	65.1	80.5	80.2
MOT20	CenterTrack	41.8	51.3	40.3
	FairMOT	54.6	61.8	67.3
	ByteTrack	61.3	77.8	75.2
	BoT-SORT	62.6	77.7	76.3
	ByteTrack*	63.4	78.1	77.2
ShipsMOT	CenterTrack	48.6	63.7	64.8
	FairMOT	54.8	67.3	70.8
	ByteTrack	66.2	70.3	79.6
	BoT-SORT	66.2	69.5	77.2
	ByteTrack*	69.2	73.6	83.4

TABLE 9 | Comparison against various detection models on ShipsMOT.

Detector	Precision↑	Recall↑	mAP50↑
CenterNet	0.929	0.902	0.891
YOLOX	0.957	0.908	0.921
YOLOv8 (baseline)	0.963	0.947	0.932
JDR-YOLO (ours)	**0.970**	**0.951**	**0.943**

Note: Bold values indicate the performance of the proposed method.

In comparison to ByteTrack, CSTrack outperforms baselines by 2.2% in HOTA and 3% in MOTA and reduces the number of ID switches from 197 to 170, primarily due to the confidence-based adaptive feature fusion (CAFF) strategy, which dynamically weights motion and appearance features based on detection confidence (Equation 10). For low-confidence detections (e.g., occluded ships), CAFF increases motion feature weight, reducing track loss by 15% compared to ByteTrack. When employing YOLOv8 as the detection model, CSTrack also surpasses other tracking methods in all metrics, which maintains high ID consistency while increasing HOTA and MOTA. Notably, the adoption of the proposed JDR-YOLO as the detection model further enhances tracking performance. CSTrack maintains its lead in overall performance and further reduces the incidence of ID switches. The results show that JDR-CSTrack not only enhances tracking accuracy but also effectively reduces ID switching errors, ensuring the continuity of tracking trajectories.

Figure 10 shows a qualitative comparison of tracking results among ByteTrack, BoT-SORT and CSTrack, all based on YOLOv8. The first column of Figure 10 shows that all MOT methods accurately track the ships in the 460th frame of the video. However, in the 530th frame, as shown in the second column of Figure 10, where the ship indicated by the blue bounding box experiences partial occlusion, ByteTrack incurs track loss, whereas both BoT-SORT and CSTrack continue to track the object successfully. When the previously occluded ship indicated by the blue bounding box reappears in the 570th frame of the video, as shown in the third column of Figure 10, both ByteTrack and BoT-SORT switch the ID of the ship, whereas CSTrack successfully maintains ID consistency. These results show that CSTrack can effectively maintain trajectory continuity and ID consistency in challenging situations such as ship occlusions.

5.3.3 | Ablation Study

We perform ablation experiments to validate the effectiveness of individual modules in JDR-CSTrack, which consists of the object detection model (JDR-YOLO) and the object tracking method (CSTrack), as detailed in Table 11 (the "" symbol denotes the adoption of a specific module of JDR-YOLO; full checkmarks indicate the use of JDR-YOLO as the detection model, whereas no checkmarks indicate the use of YOLOv8 as the detection model). Results show that the CSTrack we designed outperforms ByteTrack across all evaluated metrics. Additionally, each module in our designed JDR-YOLO also contributes to improvements in various metrics. Further analysis of the ablation study reveals distinct contributions from each component:

- *MultiCSP module*: In the backbone network, this module mitigates feature learning conflicts between detection and

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TABLE 10 | Comparison against MOT methods on ShipsMOT.

Detector	Method	HOTA↑	DetA↑	AssA↑	MOTA↑	IDF1↑	IDs↓
CenterNet	CenterTrack	48.6	44.8	55.2	63.7	64.6	203
	FairMOT	54.8	47.4	69.6	67.3	70.8	199
YOLOX	ByteTrack	66.2	58.1	77.2	69.3	79.6	197
	BoT-SORT	66.2	58.0	77.4	69.5	77.2	185
	CSTrack	68.4	61.0	78.9	72.3	81.3	170
YOLOv8 (baseline)	ByteTrack	69.5	60.8	81.9	73.6	83.4	162
	BoT-SORT	69.7	60.0	81.4	73.9	83.7	160
	CSTrack	69.7	61.0	82.0	74.1	83.8	157
JDR-YOLO (ours)	ByteTrack	70.0	61.3	81.0	76.2	85.2	147
	BoT-SORT	**70.2**	60.4	82.2	76.5	85.2	143
	CSTrack	70.1	**62.1**	**83.0**	**77.3**	**85.7**	**140**

Note: Bold values indicate the performance of the proposed method.

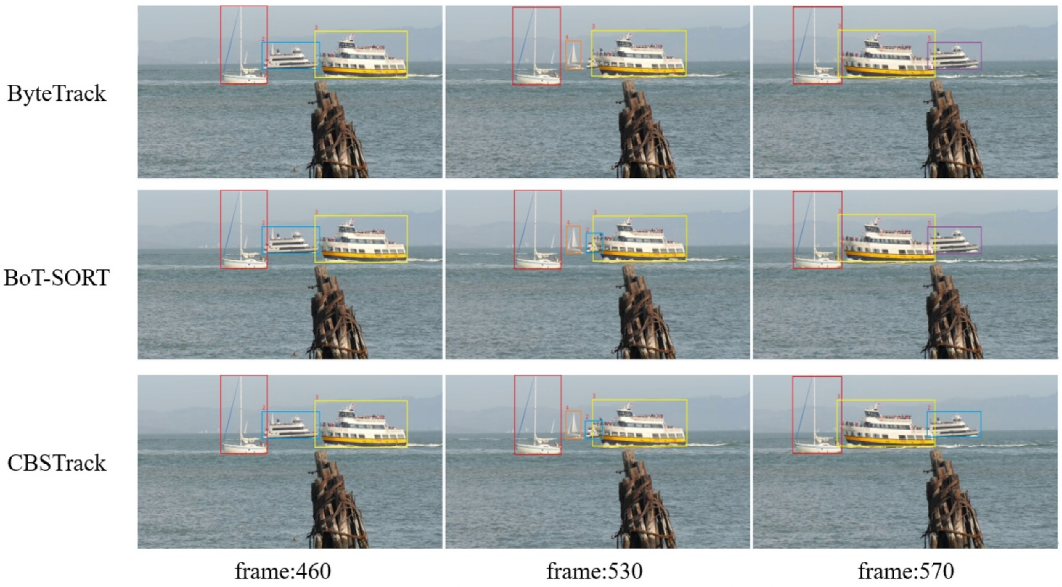


FIGURE 10 | Qualitative tracking results in a harbour occlusion scenario: (1) Frame 530: ByteTrack loses the blue ship due to partial occlusion. (2) Frame 570: BoT-SORT switches ID when the ship reappears, whereas CSTrack maintains ID consistency via the CAFF strategy.

Re-ID tasks by incorporating multibranch feature fusion, which improves MOTA by 0.5% and 0.7% on ByteTrack and CSTrack, respectively, compared to when the module is not used.

- *RFB-CA module*: This module is added in the neck network. Leveraging dilated convolutions and coordinate attention, it enhances scale robustness. Compared to the setup with no additional modules, ByteTrack incorporating this module achieves a 0.3% increase in MOTA and a reduction of six instances in ID switches. Moreover, CSTrack using this module reduces ID switches by 17 instances compared to the version without this module.
- *WIoU loss*: We added this loss function in the detection head. This component reduces misalignment errors for ships with extreme aspect ratios (e.g., barges, canoes). Compared to ByteTrack with no additional loss function, CSTrack using this loss function achieves a 0.5% increase in HOTA and a 2.3% increase in MOTA.

When all modules are integrated, the resulting model is our JDR-YOLO object detection model. Both ByteTrack and CSTrack with JDR-YOLO show a 0.4% improvement in HOTA compared to the baseline without any modules, while achieving improvements of 1.7% and 3.2% in MOTA, respectively. These enhancements contribute to improved stability in data association. Furthermore, in experimental evaluations, the CSTrack we designed demonstrates superior performance across all evaluated metrics compared to ByteTrack. Notably, the number of identity switches decreases from 152 to 140. This reduction indicates the advantage of the proposed confidence-based adaptive feature fusion association strategy in mitigating identity switches and identity losses. These results validate the effectiveness of the JDR-CSTrack framework that integrates JDR-YOLO and CSTrack.

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5.3.4 | Computational Efficiency Analysis

Real-time performance is critical for multiobject tracking tasks, as it determines practical application capabilities. To comprehensively evaluate the scalability and performance of various algorithms, we compared their parameters, memory usage and FPS across three NVIDIA GPUs: GeForce GTX 1650 4 GB, GTX 3060 6 GB and A100 40 GB, using a subset of video sequences from the ShipsMOT test set for testing and statistical analysis of average data, as shown in Table 12.

In comparison with other methods, CSTrack exhibits better stability and FPS efficiency when using YOLOv8 as the detector than JDE and CenterTrack; although using YOLOX as the detector reduces its real-time performance slightly due to larger model parameters, the performance remains within an acceptable range. The JDR-CSTrack algorithm exhibits remarkable stability and efficiency in FPS: It achieves 9.95 FPS on the GTX 1650, comparable to YOLOv8-based CSTrack and Bot-SORT, and significantly outperforms YOLOX-based trackers; on the GTX 3060, its FPS approaches that of other efficient algorithms; on the A100, it reaches 17.91 FPS, fully leveraging hardware resources to validate its applicability and stability across different platforms. Although JDR-CSTrack has a slightly lower FPS than ByteTrack, it achieves better multiobject tracking accuracy, primarily attributed to its well-designed detection and data association modules, which prioritise tracking accuracy and consistency over pure speed. This trade-off enables JDR-

TABLE 11 | Ablation study of JDR-CSTrack on ShipsMOT.

	JDR-YOLO						
Association method	**MultiCSP**	**RFB-CA**	**WIoU**	**HOTA↑**	**MOTA↑**	**IDF1↑**	**IDs↓**
ByteTrack				69.5	73.6	83.4	162
	✓			69.6	74.1	83.5	157
		✓		69.6	73.9	84.0	156
			✓	69.9	74.6	84.3	151
	✓	✓		69.8	74.5	84.2	154
	✓		✓	69.7	74.2	83.9	158
		✓	✓	69.7	74.9	83.6	156
	✓	✓	✓	69.9	75.3	84.2	152
CSTrack				69.7	74.1	83.8	157
	✓			69.8	74.8	84.5	151
		✓		69.7	74.1	84.2	**140**
			✓	70.0	75.9	84.7	150
	✓	✓		69.9	75.0	84.7	145
	✓		✓	**70.1**	77.0	85.6	142
		✓	✓	70.0	76.3	85.2	141
	✓	✓	✓	**70.1**	**77.3**	**85.7**	**140**

Note: Bold values indicate the performance of the proposed method.

TABLE 12 | Real-time comparison of different multiobject tracking algorithms.

		Memory used (MB)			FPS (avg.)		
Method	**Parameters (M)**	**GTX 1650**	**GTX 3060**	**A100**	**GTX 1650**	**GTX 3060**	**A100**
JDE	23.40	370.32	514.74	700.05	7.75	11.11	13.67
CenterTrack	25.60	364.95	507.28	699.90	5.50	8.26	10.16
FairMOT	25.60	366.10	508.88	692.08	10.98	15.21	18.71
ByteTrack_YOLOX	52.20	403.57	964.98	1312.94	7.26	10.49	12.90
ByteTrack_YOLOv8	43.69	373.20	893.50	1215.68	14.98	20.33	25.01
Bot-SORT_YOLOX	52.20	554.64	1071.77	1424.45	3.13	5.41	6.65
Bot-SORT_YOLOv8	43.69	478.58	923.94	1227.97	9.73	14.04	19.04
CSTrack_YOLOX	52.20	581.10	1082.38	1456.11	4.52	7.01	8.62
CSTrack_YOLOv8	43.69	484.14	909.56	1223.62	10.74	14.96	19.71
JDR-CSTrack	45.66	484.45	916.62	1209.41	9.95	14.33	17.91

Note: Bold values indicate the performance of the proposed method.

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CSTrack to maintain superior object association and occlusion handling capabilities, making it an ideal choice for scenarios where tracking precision is more critical than real-time performance.

In conclusion, JDR-CSTrack excels in overall stability and low memory consumption, positioning it as a candidate for deployment on low-performance devices while ensuring robust real-time multiobject tracking capabilities.

6 | Conclusion

In this work, we focus on multiobject tracking for ships. We build a new ship multiobject tracking (ShipsMOT) dataset, which provides high-resolution and diverse ship videos, along with more object trajectories. We observed that existing MOT methods are primarily designed for general objects and are not well-suited for ship objects, which can limit their performance in this specific task. To address this, we propose a new framework, JDR-CSTrack, specially designed to enhance the feature extraction capability for ship objects and to improve tracking performance under challenging conditions such as occlusion, object similarity and large-scale differences. However, ShipsMOT also has some imbalance. For one aspect, there is scenario imbalance, as it lacks video images captured under severe weather conditions such as rainstorm and typhoon. For another, due to privacy issues, warship-related videos are hard to collect, resulting in category imbalance for specific types of ships. Therefore, in our future work, we will consider exploring ship multiobject tracking in more real-world scenarios and employ methods such as data augmentation to address the current limitations, further promoting the development of ship detection and MOT in practical applications.

Author Contributions

Fang Luo: conceptualization, funding acquisition, project administration, supervision, validation, writing – original draft. **Pengju Jiang:** data curation, formal analysis, methodology, software, validation, writing – original draft, writing – review and editing. **George To Sum Ho:** formal analysis, supervision, validation, writing – review and editing. **Wenjing Zeng:** validation, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study can be downloaded from https://github.com/jpj0916/ShipsMOT.

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